

Distributed Cooperative Steering Control for Three-Section Six-Axis Articulated Vehicles

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Abstract—This paper proposes a distributed cooperative control strategy to address the accuracy of trajectory following control of six-axle articulated vehicles with three-section formation. Leveraging a kinematic-derived dynamic model, the strategy employs combined Extended Ackermann feedforward and Fuzzy PID control for the 4th and 6th axles, mitigating large lateral deviations via dynamic PID parameter optimization. Standard Fuzzy PID controls the 2nd, 3rd, and 5th axles. Co-simulation results using TruckSim and Simulink demonstrate significant reductions in critical axle lateral deviations compared to conventional PID, alongside effective management of other axles under typical road conditions. The proposed approach exhibits high precision and robustness, offering an effective solution.

Keywords—Articulated vehicle, Fuzzy PID control, Trajectory tracking, Multi-axle cooperative control

I. INTRODUCTION

With the acceleration of urbanization and the increasing demand for public transportation, efficient and environmentally friendly means of transportation has become an important support for sustainable urban development [1-2]. As a useful supplement to the main urban public transportation system such as ordinary buses, subways and streetcars, six-axle articulated vehicles have gradually become an important part of the urban public transportation system by virtue of their large capacity, low energy consumption and flexible turning performance. This type of vehicle consists of multiple compartments connected by articulated devices, with multiple axial degrees of freedom of movement, and is capable of realizing efficient transportation in complex urban road environments. However, their unique structural design and multi-axis motion characteristics also bring unprecedented challenges to the development of control systems. Especially in realizing high-precision trajectory following control, traditional control methods are often difficult to meet its high requirements for stability and accuracy. Therefore, the study of multi-axis cooperative control strategies for six-axis articulated vehicles to realize accurate trajectory following control not only has important theoretical significance, but also shows a

broad application prospect.

Trajectory following control is one of the core technologies in automatic driving and intelligent transportation systems, and its goal is to enable the vehicle to follow a predetermined path while maintaining a stable attitude and speed. For a six-axis articulated bus vehicle, due to its multi-axis articulated structure, the coordinated motion between the axes is especially critical in the driving process. Traditional trajectory following control methods are mostly designed for single-axis vehicles or simplified models, which are difficult to be directly applied to the control requirements of such complex systems. In recent years, with the rapid development of control theory and computational technology, advanced control strategies based on model predictive control, sliding mode control, fuzzy control and other advanced control strategies are increasingly widely used in vehicle trajectory following control, which provide new theoretical support and technical paths for the control system design of six-axis articulated bus vehicles. Dasol Jeong [3] et al. designed a pipelined robust model predictive control, which is effectively applied to the control system of six-axis articulated bus vehicles through pipeline analysis and constraint tightening techniques to effectively deal with modeling errors and uncertainties, and to achieve high-precision trajectory following control of a 3-axis articulated vehicle. Mateusz Szumilas [4] et al. focused on the joint control of articulated buses, and adjusted the damping of the joints through the use of MEMS sensors and electro-hydraulic servo systems to ensure the stability and maneuverability of the vehicle at different speeds. Rui Li [5] et al. proposed a trajectory following algorithm based on adaptive sliding mode control for articulated vehicle control. By verifying in a hardware-in-the-loop simulation environment, the ASMC method is able to effectively control the lateral position deviation, heading angle deviation and curvature deviation, and it reduces the errors and oscillations and improves the following performance compared with the traditional method. Peng Jing [6] et al. proposed an autonomous guidance and trajectory following control system, which realizes the path following of the Smart Tram by using the method of sectional control. Li Si Xu [7] et al. proposed a multi-point pre-scanning path following method for articulated vehicles based on predictive control of dynamics model, which

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realizes high-precision path following control of articulated vehicles by constructing a dynamics model and adopting the control objective of multi-point pre-scanning error. Meanwhile, the significant improvement of sensor technology (e.g., LIDAR, vision system) and real-time computing capability makes the vehicle able to perceive the complex environment in real time and make decisions quickly, which further promotes the research and practice of high-precision trajectory following control.

Although the research on trajectory following control of six-axis articulated bus vehicles has made progress, it still faces challenges. Its multi-axis articulated structure leads to highly nonlinear kinematics and dynamics model, which increases the complexity of control system design. In addition, uncertainties such as road surface unevenness, side winds, and dynamic traffic place higher demands on the robustness of the control system. Therefore, the development of efficient and reliable multi-axis cooperative control strategies to realize high-precision trajectory following control is a key problem to be solved nowadays.

In this paper, a novel distributed hybrid control architecture is proposed for a three-section six-axis vehicle. This architecture judiciously assigns a Combined Fuzzy PID (CFPID) controller to the critical 4th and 6th axles and a standard Fuzzy PID (FPID) controller to the 2nd, 3rd, and 5th axles, balancing control performance and computational efficiency. Based on the real vehicle data provided by the project of CRSC Corporation, a co-simulation environment integrating TruckSim and Simulink is established to validate the control scheme under realistic urban road conditions. The proposed control architecture significantly outperforms the conventional PID method in terms of lateral deviation minimization, especially on critical axles.

Specifically, this study will first establish a dynamic model of a six-axis articulated vehicle and analyze its motion behavior under different working conditions, and then design a distributed multi-axis cooperative control strategy to achieve smooth driving and accurate trajectory following through coordinated allocation of the cornering control of each axle, and finally, verify the validity and robustness of the proposed method through simulation experiments.

II. VEHICLE DYNAMICS MODELING

The experimental simulation vehicle is a three-compartment, six-axle articulated bus with power centralized drive, the front axle is the drive axle, and all axles can be steered. The structure diagram of the articulated vehicle is shown in Fig. 1, where A_i ($i=1,2,\dots,6$) denotes the axle number, δ_i ($i=1,2,\dots,6$) denotes the angle of rotation of each axle during the driving process, and O_i ($i=1,2,3$) denotes the center-of-mass point of each compartment. The mass of the trailer $m_1=9000\text{kg}$, the mass of the intermediate trailer $m_2=6000\text{kg}$, and the mass of the trailing trailer $m_3=9000\text{kg}$. The distances from the center of mass of the trailer to the front axle and to the rear axle are denoted by $a_1=2550\text{mm}$ and $b_1=4150\text{mm}$, respectively, and similarly the distances from the center of mass of the intermediate trailer to the front and rear axles are denoted by $a_2=3350\text{mm}$ and $b_2=3350\text{mm}$, while the distances from the

center of mass of the trailing trailer to the front and rear axles are denoted by $a_3=4150\text{mm}$ and $b_3=4150\text{mm}$. $c_1=4800\text{mm}$ and $d_1=5500\text{mm}$ are used to indicate the length from the center of mass of the tractor to the head and tail of the carriage, respectively, and similarly the lengths from the centers of mass of the last two trailers to the head and tail of the carriage are indicated by $c_2=4700\text{mm}$, $c_3=5500\text{mm}$ and $d_2=4700\text{mm}$, $d_3=4800\text{mm}$, respectively.

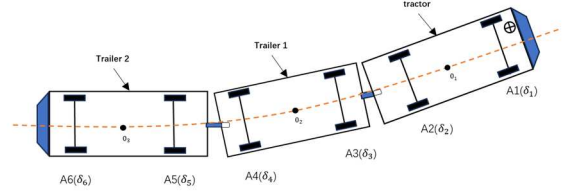
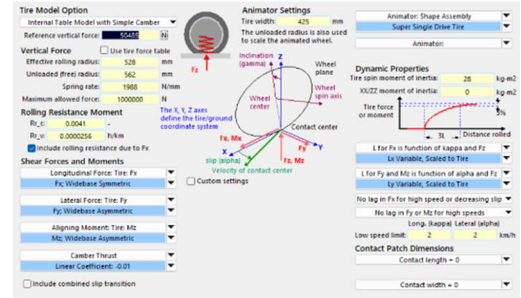


Fig. 1. Articulated Vehicle Structure



(a)



(b)

Fig. 2. Trucksim Vehicle And Tire Model

To describe the vehicle behavior with sufficient fidelity, the dynamic model is established using the TruckSim platform. The articulated structure is modeled as a multi-body system consisting of a tractor and two trailer units, connected via rotary joints. Each axle is represented with suspension and steering parameters mapped from the actual vehicle configuration. The front body is modeled as a 2-axle bus with power input and steering capabilities. The trailers are configured as 2-axle passive units. The current model utilizes TruckSim's built-in dynamics solver, and its structure corresponds to a simplified 9-DOF transverse model involving transverse displacement, yaw angle, and articulation angle. An example simplified order dynamic model is described below:

$$M\dot{v}_y + Mu\dot{\psi} = \sum F_{yi} l_z \dot{\omega}_z = \sum M_{zi}$$

where the lateral tire forces and yaw moments. These are

handled indirectly by TruckSim's in-house multi-body simulation and have been validated in previous studies. A vehicle dynamics model and tire model data is shown Fig. 2 can be constructed in TruckSim software. The TruckSim vehicle model has been parameterized according to a real three-section articulated bus configuration provided by CRSC Group. Although full-scale physical testing was not conducted, the TruckSim-based multi-body model has been validated in previous literature [8,10] and widely adopted in commercial vehicle control research. Future work will include hardware-in-the-loop (HIL) testing for further verification.

III. VEHICLE CONTROLLER DESIGN

A. Wheel angle calculation based on extended Ackermann steering

The first axle steering of the vehicle is controlled by the driver during traveling, the first axle wheel steering angle δ_1 is known. In order to achieve high maneuverability and following performance, Wagner [9] et al. designed an Ackermann steering angle based calculation method, which can calculate the feedforward steering angles of A_4 and A_6 axes from the known A_1 axle steering angle. The method ensures that the rear axes (A_4 and A_6) move along the same circle described by the front axis (A_1) by basing on the classical Ackermann steering principle, which allows the instantaneous center of rotation P , to be obtained from the steering angle of the front axis (A_1). Specifically, the steering angles of the rear axes are deduced from geometrical relations such that the centroids of all axes always lie on the same circle.

As shown in Fig. 3, the radius of rotation of the A_1 axis is R_1 . Assuming that the radius of rotation of the A_2 and A_6 axes are equal to that of the A_1 axis, there is a qualification:

$$R_1 = R_2 = R_6 \quad (1)$$

The radius of rotation of the A_2 axis is obtained from the geometric relationship:

$$R_2 = \frac{L_1}{\tan \delta_1} \quad (2)$$

Where $L_1 = a_1 + b_1$ indicates the wheelbase of the first vehicle.

The length from the first articulation point to the center of rotation is:

$$R_{g1}^2 = L_{g1}^2 + R_2^2. \quad (3)$$

In the triangle $A_4 - P - G_1$ of Fig. 2, the magnitude of the angle β_1 can be calculated from the knowledge of trigonometry:

$$\beta_1 = \arccos\left(\frac{R_{g1}^2 + L_2^2 - R_2^2}{2R_{g1}L_2}\right) \quad (4)$$

where $L_2 = c_2 + b_2$ indicates the distance from the first articulation point to the A_4 axis.

by the equation:

$$\varphi_4 = \delta_4 + \frac{\pi}{2} \quad (5)$$

As well as the math of congruent triangles, one can

calculate the size of the final A_4 axis control angle δ_4

$$\delta_{4f} = \arccos\left(\frac{R_{g1}}{R_2} \sin \beta_1\right) \quad (6)$$

The radius of motion of the second hinge through the corner δ_4 can be calculated by the following equation:

$$\begin{aligned} R_{g2} &= l_{g2}^2 + R_4^2 - 2L_{g2}R_4 \cos\left(\frac{\pi}{2} - \delta_{4f}\right) \\ &= l_{g2}^2 + R_4^2 - 2L_{g2}R_4 \sin \delta_{4f} \end{aligned} \quad (7)$$

Similarly, the magnitude of the angle of rotation δ_6 of the axis A_6 can be computed

$$\beta_2 = \arccos\left(\frac{R_{g2}^2 + L_3^2 - R_2^2}{2R_{g2}L_3}\right) \quad (8)$$

$$\delta_{6f} = \arccos\left(\frac{R_{g2}}{R_2} \sin \beta_2\right) \quad (9)$$

where $L_3 = c_3 + a_3$ denotes the distance from the rear axle of the tail car to the second articulation point.

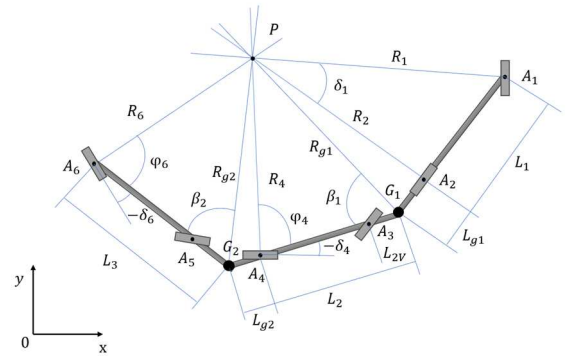


Fig. 3. Schematic Of The Kinematic Model With A Single Center Of Rotation

Assuming a shared rotation center, steering angles for the A_4 and A_6 axes can be calculated using a geometric method to provide feedforward control for vehicle trajectory following. This approach offers computational simplicity but is primarily applicable to steady-state turning conditions due to its underlying geometric assumptions. Critically, this method lacks a feedback mechanism for correcting path following errors. Consequently, this feedforward strategy necessitates further optimization to effectively handle dynamic conditions and enhance accuracy.

B. Design of control strategy based on fuzzy PID

Traditional regulators—whether PID, LQR, or sliding-mode schemes—tend to falter when confronted with the pronounced nonlinearities inherent to multi-unit vehicles; their fidelity hinges on precise plant models or they become fragile in the face of sensor noise. Fuzzy inference, by comparison, translates linguistic rules into control actions, thereby accommodating uncertain parameters and strong nonlinear couplings without recourse to an exact mathematical description. A hybrid fuzzy-PID architecture is consequently adopted to exploit this resilience, allowing gains to be reshaped on-line as road friction, load distribution, or articulation angles

vary. This design yields higher robustness, faster transients, and no need for exact models; the fuzzy layer retunes PID gains on the fly to keep lateral error low through sharp bends or uneven terrain.

In this section, a wheel cornering controller based on a fuzzy PID control strategy is proposed. This controller reduces the trajectory following deviation between the actual and desired paths of the vehicle by real-time adjustment of the vehicle's remaining axle turning angles except for the A_1 axle. The specific control process is shown in the control flow diagram in Fig. 4.

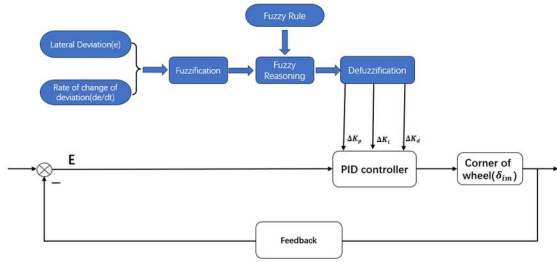
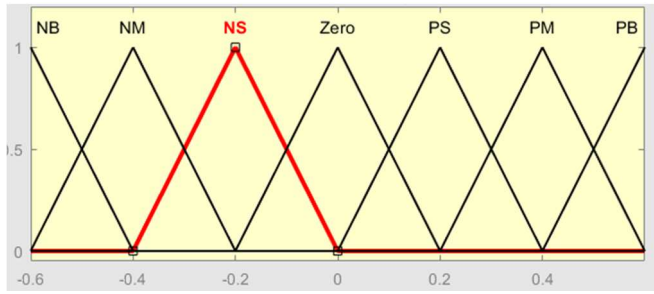
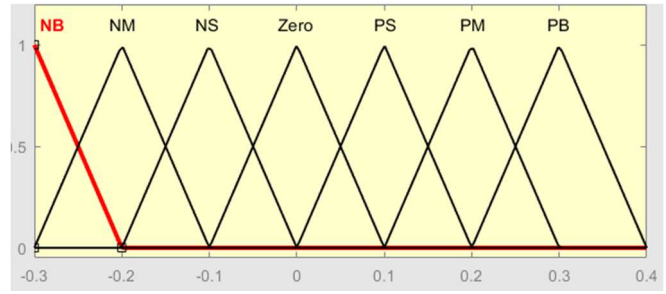


Fig. 4. Fuzzy PID Control Flow Chart



(a) Lateral deviation and deviation change rate affiliation function



1) Design of the lateral deviation calculation module

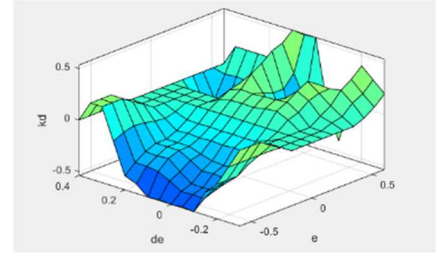
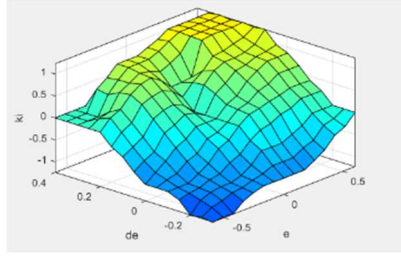
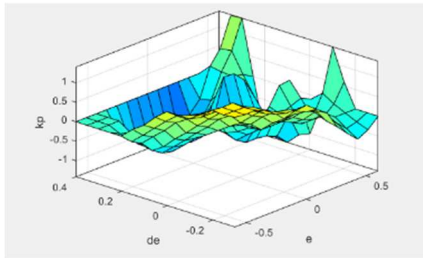
The trajectory of the A_1 axis is the expected trajectory while the trajectory of the rear axle needs to be as close as possible to the first axle's trajectory, which requires that the current trajectory data of the rear axle is obtained and the expected trajectory is calculated. Before calculating the trajectory deviation of the rear axle, it is necessary to discretize the driving trajectories of each axis into a point cloud and store it in the memory. There is a delay time between the current trajectory and the expected trajectory, which can be calculated by the delay module in Simulink [10]. The time size of the delay is:

$$\int_{t-\Delta t_{dn}(t)}^t \frac{1}{\Delta t_n(\tau)} d\tau = 1 \quad (10)$$

It can also be used in practical engineering applications by

$$\Delta t_{dn}(t) = \frac{L_n}{v} \quad (11)$$

to abbreviate the calculation, where L_n denotes the wheelbase from the first axis to the A_n axis and v denotes the vehicle speed.



(b) Joint control surfaces

Fig. 5. Affinity Function And Joint Control Surface Map

TABLE I. ΔK_p , ΔK_i AND ΔK_d RULE SHEET

e /ec	NB	NM	NS	ZO	PS	PM	PB
NB	PB/NB/PS	PB/NB/NS	PM/NM/NB	PM/NMNB	PS/NSNB	ZO/ZO/NM	ZO/ZO/PS
NM	PB/NB/PS	PB/NM/NS	PM/NS/NB	PS/ZO/NB	ZO/PS/NS	NS/PS/NM	NM/PS/ZO
NS	PM/NM/ZO	PM/NM/ZO	PS/NS/NS	ZO/ZO/NS	NS/PS/NS	NM/PM/NS	NM/PM/ZO
ZO	PM/NM/ZO	PS/NS/ZO	ZO/ZO/PS	ZO/ZO/PS	ZO/ZO/PS	NS/ZO/PS	NM/PM/ZO
PS	PS/NM/ZO	PS/NM/ZO	ZO/NS/ZO	NS/ZO/PS	NM/PM/PS	NM/PM/PS	NB/PB/ZO
PM	PS/NS/ZO	ZO/NS/ZO	NS/ZO/ZO	NM/PS/PS	NM/PM/PM	NB/PB/PS	NB/PB/ZO
PB	ZO/ZO/ZO	ZO/ZO/PM	NM/PS/PB	NM/PM/PB	NB/PB/PB	NB/PB/PS	NB/PB/PS

The coordinates of the posterior axis $p_n(x_n(t), y_n(t))$ ($n=2,3,4,5,6$) can be obtained from this delay, where $x_n(t), y_n(t)$ are:

$$\begin{cases} x_n(t) = x(t - \Delta t_{dn}(t)) \\ y_n(t) = y(t - \Delta t_{dn}(t)) \end{cases} \quad (12)$$

After obtaining the coordinates of the rear axes, the trajectory deviation of the rear axes and the rate of change of the deviation can be calculated.

$$dx = (x - x_n) \quad (13)$$

$$dy = (y - y_n) \quad (14)$$

$$e = \sqrt{(dx)^2 + (dy)^2} \quad (15)$$

$$\dot{e} = \frac{de}{dt} \quad (16)$$

2) Fuzzy rules and affiliation function design

The procedure for defining fuzzy control rules follows a structured expert-based approach. Initially, extensive vehicle simulation data were analyzed to determine the typical patterns of lateral deviation and its rate of change during trajectory following. Based on these observations, the fuzzy rules were formulated by domain experts to reflect intuitive corrective behaviors under different driving scenarios. For example, large negative deviation combined with an increasing error rate requires aggressive corrective steering, which is encoded as high gain increases in the fuzzy PID controller.

The membership functions were selected as symmetric triangular functions to ensure real-time computation efficiency and simplicity of tuning and the joint control surface and affiliation function [11] are shown in Fig. 5. Their ranges were calibrated based on statistical bounds observed in simulation results and matched to control variable domains used in prior fuzzy control studies [12]. This hybrid method—combining expert knowledge with empirical data fitting—ensures that the fuzzy controller is well-tuned to the articulated vehicle's

nonlinear characteristics while remaining generalizable to a wide range of road conditions.

The fuzzy PID controller designed in this paper has two inputs-lateral deviation 'e' and its rate 'ec' -and three outputs- ΔK_p , ΔK_i and ΔK_d . The domains of the input variables are [-0.6 0.6] and [-0.3 0.4], and the domains of the output variables are [-1.6 1.6], [-1.4 1.4], and [-0.6 0.6], respectively. The fuzzy sets of all the input and output variables are set to {NB NM NS ZO PS PM PB}. The output variables ΔK_p , ΔK_i and ΔK_d , and the specific rules are shown in Tables 2 to 4. The improved PID parameters by fuzzy reasoning are:

$$\begin{cases} K_p = K_{p0} + \Delta K_p \\ K_i = K_{i0} + \Delta K_i \\ K_d = K_{d0} + \Delta K_d \end{cases} \quad (17)$$

Where K_p , K_i , K_d and K_{p0} , K_{i0} , K_{d0} are the final and initial values of the PID coefficients.

C. Control Policy Assignment

Vehicles are susceptible to a variety of disturbing factors during turn-by-turn driving, which may exacerbate the trajectory deviation of vehicles. In order to enhance the robustness of three-knuckle six-axle intelligent rail train in trajectory following deviation control, this paper proposes an optimized control strategy allocation scheme (distributed cooperative control). In this study, a distributed cooperative control strategy is proposed to enhance the trajectory tracking accuracy of six-axle articulated vehicles. Unlike centralized steering control, where a central controller governs all axle angles, the distributed control framework adopts a modular approach. Each axle or axle group is equipped with an independent local controller that receives coordination signals from a supervisory controller. These local controllers execute control tasks (e.g., fuzzy PID adjustment or feedforward compensation) based on their respective lateral deviation inputs. The key advantage of this framework lies in its fault

tolerance, scalability, and robustness against localized disturbances. A schematic diagram of the distributed framework is shown in Fig.6, illustrating the hierarchical structure of coordination.

In the previous analysis, this paper identifies that the most significant deviation of trajectory deviation in the three-knuckle intelligent rail train is the A_4 and A_6 axes, so for the control of these two axes, this paper adopts the feed-forward control superimposed on fuzzy PID with extended Ackermann's turn angle (abbreviated as CFPID) [10]. The specific control process is as follows: firstly, the feedforward controller calculates the feedforward angles δ_{4f} and δ_{6f} of A_4 and A_6 axes through the vehicle parameters and the angle of A_1 axes, and then calculates the trajectory deviation and the deviation change rate of A_4 and A_6 axes through real-time detection during driving, and passes these parameters into the fuzzy controller. The fuzzy controller dynamically adjusts the gain parameters of the PID controller based on the fuzzy inference rules so as to optimize the PID parameters in real time, thus obtaining the regulated wheel steering angles δ_{4m} and δ_{6m} for the axes A_4 and A_6 . The final wheel steering angles for the two axes are:

$$\begin{cases} \delta_4 = \delta_{4f} + \delta_{4m} \\ \delta_6 = \delta_{6f} + \delta_{6m} \end{cases} \quad (18)$$

For the control of A_2 , A_3 and A_5 axes, this paper directly adopts the fuzzy PID controller to calculate their wheel turning angles. Specifically, by detecting the trajectory deviation of each axis and its change rate in real time, the fuzzy PID controller dynamically adjusts the PID parameters based on the fuzzy inference rules, and finally outputs the corresponding wheel turning angles, which are noted as δ_{2m} , δ_{3m} , and δ_{5m} , respectively, and these values are the final wheel turning angles- $\delta_2, \delta_3, \delta_5$.

In this distributed control architecture, axles 4 and 6 use a hybrid feedforward + fuzzy PID controller, while axles 2, 3, and 5 apply standard fuzzy PID. All controllers operate independently, coordinated by a supervisory layer

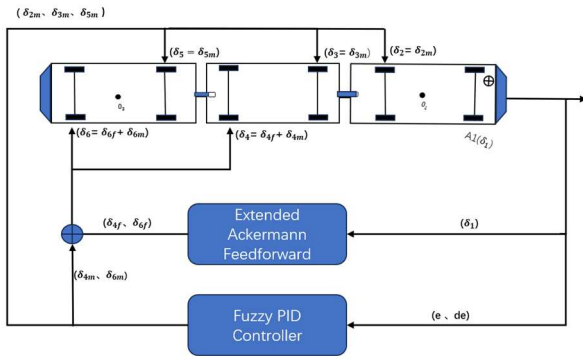


Fig. 6. Distributed Collaborative Control Flowchart

IV. SIMULATION EXPERIMENT ANALYSIS

A. Simulation Experiment

In order to verify the performance of the designed controller, this study builds a vehicle model in Truksim and co-simulates

it with Simulink. In the simulation environment designed in this paper, the road model includes three straight lines with lengths of 10m, 40m and 30m respectively and two 90° curves with radii of 25m and 15m respectively, and the specific layout is shown in Figure 7. In addition, considering the influence of the road friction coefficient on the dynamic behavior of the vehicle, this paper sets it to the dry asphalt road adhesion coefficient $\mu = 0.85$ [13]. At the same time, considering that too fast turning speed may cause risks such as vehicle rollover, the simulation speed is set to 10km/h.

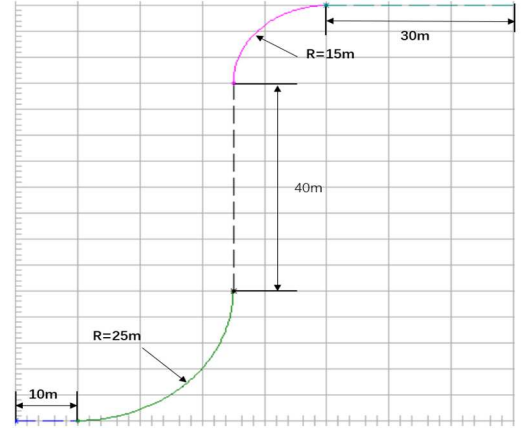
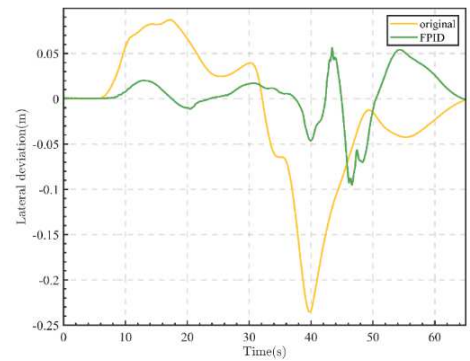


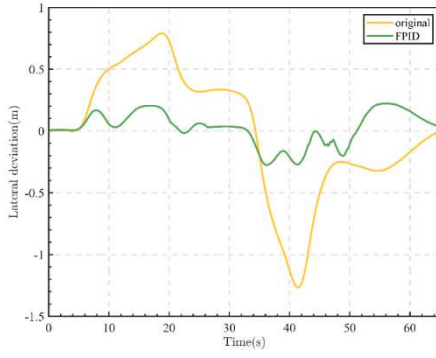
Fig. 7. Simulation Roadmap

B. Simulation results analysis

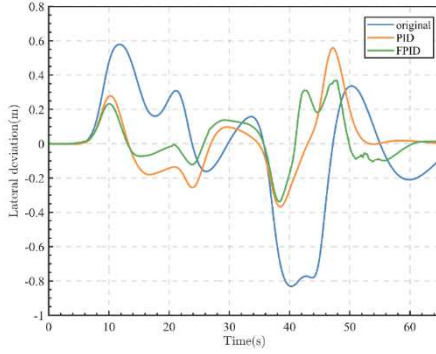
According to the control strategy and simulation environment designed in the previous section, the simulation results of the lateral deviation of vehicle $A_1 \sim A_6$ axes are obtained through the joint simulation of Simulink and Truksim, as shown in Fig. 8. The 'original' curve represents the uncontrolled case, where only the first axle is steered according to the path and no corrective steering control is applied to the subsequent axes (2 through 6). This serves as a baseline to demonstrate the magnitude of the inherent trajectory deviation.



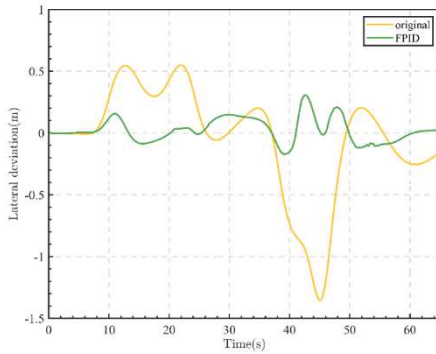
(a) A_2 axis transverse deviation



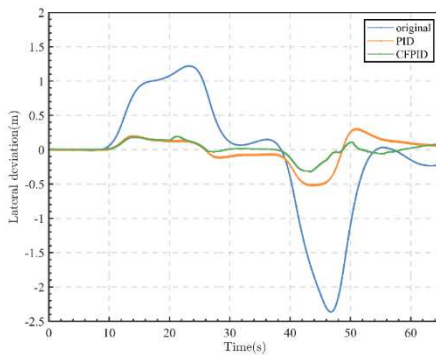
(b) A_3 axis transverse deviation



(c) A_4 axis transverse deviation



(d) A_5 axis transverse deviation



(e) A_6 axis transverse deviation

Fig. 8. $A_2 \sim A_6$ Axis Lateral Deviation Simulation Results

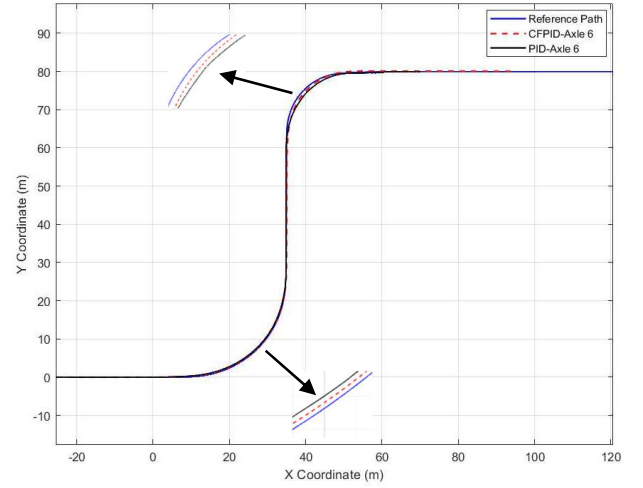


Fig. 9. A_6 Axis Trajectory Tracking Result

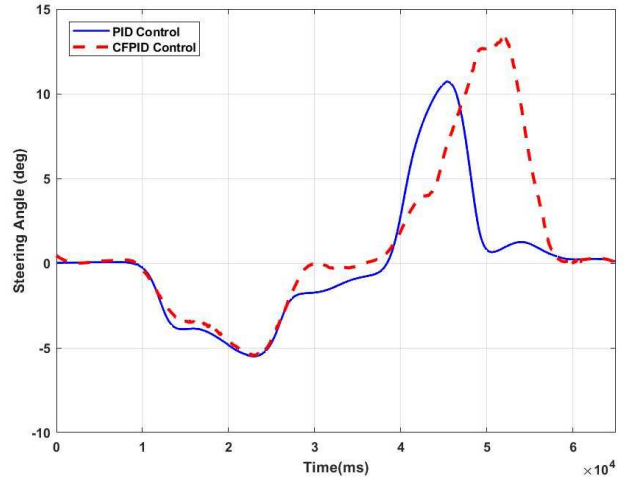


Fig. 10. A_6 Axis Steering angle

The simulation experiments show that the maximum values of the lateral deviation results of A_4 and A_6 axes in the uncontrolled state are 0.84m and 2.4m, respectively, and the lateral deviations of the two axes are reduced to within 0.37m and 0.32m, respectively, under the effect of the compensating fuzzy PID control; the maximum values of the lateral deviations of the three axes in the uncontrolled state are 0.24m, 1.32m and 1.38m for the three axes in the uncontrolled state, and the maximum values are 0.058m, 0.22m and 0.33m for the three axes after the application of the fuzzy PID control, 1.32m and 1.38m, respectively, and the lateral deviations of the three axes are reduced to within 0.058m, 0.22m and 0.33m, respectively, after applying the fuzzy PID control.

In order to reflect the control performance of this control strategy, this paper also carries out the PID control test on A_4 and A_6 axes, and the simulation results show that the maximum lateral deviation of the two axes under PID control is 0.57m and 0.51m respectively, which further verifies that the control strategy designed herein is superior to the performance

of traditional PID. After applying the proposed control, the maximum lateral deviations for all controlled axles are significantly reduced, with values ranging from 0.096m (for A2) to 0.37m (for A4), as detailed in Table II.

Figure 9 shows the trajectory tracking results for the 6th axle under both a standard PID controller and the proposed CFPID controller. The reference path is a double lane change maneuver. Both controllers successfully guide the vehicle along the path, but a clear performance difference is evident. The PID controller's trajectory (solid black line) shows a visible deviation from the reference path, particularly as it enters and exits the S-curve. In comparison, the path of the proposed CFPID controller (dashed red line) maintains a much closer alignment with the reference path, demonstrating significantly higher accuracy.

This enhanced tracking performance corresponds to the improved control action shown in Figure 10.

TABLE II. *Simulation results*

axles	Control methods	Maximum lateral deviation (m)
A_2	FPID	0.096
A_3	FPID	0.28
A_4	CFPID	0.37
A_5	FPID	0.35
A_6	CFPID	0.32

V. Conclusions

In this paper, a distributed cooperative control strategy is successfully designed and validated for the control difficulties faced by three-unit six-axis articulated vehicles in complex trajectory following. The core innovation of the strategy is that the extended Ackermann feed-forward control is integrated to predict the ideal corner for the 4th and 6th axes, which have significant lateral deviations, and combined with a fuzzy PID controller for dynamic error compensation; at the same time, the standard fuzzy PID control is applied to the 2nd, 3rd, and 5th axes. The key research results are confirmed by the joint simulation of TruckSim and Simulink: compared with the traditional PID control, the proposed synergistic strategy significantly improves the trajectory following accuracy of the key articulated axes, effectively suppresses the lateral deviation, and takes into account the stability of each part of the vehicle. This study not only verifies the feasibility and superiority of the distributed hybrid control architecture, but also provides a validated control scheme with good robustness and application prospects for solving the high-precision path navigation problem of similar long, multi-unit articulated vehicles.

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